

Challenge (Habanero)

Open Ended Questions (FRQ)

- Q9.** A conservation group wants to protect a species of bird by creating a wildlife reserve. They have 10,000 to spend on food, habitat preparation per month, habitat preparation costs 2000 per month, and staffing costs 1500 per month. If they want to provide food for at least 5 months, prepare habitat for at least 2 months, and hire staff for at least 3 months, determine the maximum amount they can spend on each category. Show your work and provide a feasible solution.
- Q10.** A park ranger wants to create a hiking trail with a maximum length of 10 miles. The trail has two sections: one with a steep incline and one with a gentle decline. The steep incline section can be at most 3 miles long and the gentle decline section can be at least 4 miles long. If the ranger wants to allocate the trail length to minimize the steep incline section, determine the maximum length of the gentle decline section. Show your work and provide a feasible solution.

Chef's Tips

- Emphasize the importance of reading and understanding the problem before starting to solve it.
- Encourage students to use visualization techniques to understand the constraints and feasible solutions.
- Remind students to check their units and ensure they are consistent throughout the problem.